Size & Shape, staining intensity, and staining texture of hematoxylin (NH) or eosin (NE)

Feature symbol 1	Feature symbol 2	Feature Name	Feature Type	Origin	Paper/Website	Additional Info	Remarks
N_F1		Solidity	Shape features (by Matlab's regionprops)	Matlab	https://www.mathworks.com/help/images/ref/regionprops.html		
N_F2		Extent					
N_F3		EquivDiameter					
N_F4		Eccentricity					
N_F5		MinorAxisLength/MajorAxisLength					
N_F6		Area					
N_F7		Perimeter					
NH_F8	NE_F8	Mean	Intensity histogram based metrics:	Matlab implementation (code adjusted for 2D) PORTSFeat(1:6)	https://www.mathworks.com/matlabcentral/fileexchange/55587-ports-3d-image-texture-metric-calculation-package (see example) function: compute_histogram_metrics	https://pyradiomics.readthedocs.io/en/latest/features.html#module-radiomics.glrlm	
NH_F9	NE_F9	Variance					
NH_F10	NE_F10	Skewness					
NH_F11	NE_F11	Kurtosis					
NH_F12	NE_F12	Energy					
NH_F13	NE_F13	Entropy					
NH_F14	NE_F14	Angular second moment	Features of Gray-Tone-Spatial-Dependence-Matrix (GTSDM)	PORTSFeat(7:26)	function: compute_3D_GTSDM		called "Energy" in Soh 1999
NH_F15	NE_F15	Contrast					
NH_F16	NE_F16	Correlation					
NH_F17	NE_F17	Sum of squares variance					
NH_F18	NE_F18	Inverse Difference moment				van Griethuysen et. al., 2019	called "Homogeneity" in Soh 1999
NH_F19	NE_F19	Sum average					
NH_F20	NE_F20	Sum variance					
NH_F21	NE_F21	Sum Entropy					
NH_F22	NE_F22	Entropy					
NH_F23	NE_F23	Difference Variance					
NH_F24	NE_F24	Difference Entropy					
NH_F25	NE_F25	Information Correlation 1					
NH_F26	NE_F26	Information Correlation 2					
NH_F27	NE_F27	Maximal Correlation Coefficient					*** NOT COMPUTED -- ALWAYS ZERO ***
NH_F28	NE_F28	Autocorrelation	Texture Features from Soh (1999)				
NH_F29	NE_F29	Dissimilarity					
NH_F30	NE_F30	Cluster Shade					
NH_F31	NE_F31	Cluster Prominence					
NH_F32	NE_F32	Maximum Probability					
NH_F33	NE_F33	Inverse Difference	Texture feature from Clausi (2002):			van Griethuysen et. al., 2019	
NH_F34	NE_F34	Coarseness	Features of Neighborhood-Gray-Tone-Difference-Matrix (NGTDM)	PORTSFeat(27:31)	function: compute_NGTDM_metrics		
NH_F35	NE_F35	Contrast					
NH_F36	NE_F36	Busyness					
NH_F37	NE_F37	Complexity					

NH_F38	NE_F38	Texture Strength					
NH_F39	NE_F39	Small Zone Size Emphasis	Gray-Level Zone Size Metrics (GLZSM)	PORTSFeat(32:42)	function: compute_zone_size_metrics		
NH_F40	NE_F40	Large Zone Size Emphasis				van Griethuysen et. al., 2019	
NH_F41	NE_F41	Low Gray-Level Zone Emphasis					
NH_F42	NE_F42	High Gray-Level Zone Emphasis					
NH_F43	NE_F43	Small Zone / Low Gray Emphasis					
NH_F44	NE_F44	Small Zone / High Gray Emphasis					
NH_F45	NE_F45	Large Zone / Low Gray Emphasis					
NH_F46	NE_F46	Large Zone / High Gray Emphasis				van Griethuysen et. al., 2019	
NH_F47	NE_F47	Gray-Level Non-Uniformity					
NH_F48	NE_F48	Zone Size Non-Uniformity					
NH_F49	NE_F49	Zone Size Percentage				van Griethuysen et. al., 2019	
NH_F50	NE_F50	SHORT RUN EMPHASIS (SRE)	Gray Level Run Length Features (GRLE)	Matlab implementation (code adjusted)	https://www.mathworks.com/matlabcentral/fileexchange/52640-gray-level-run-length-image-statistics		
NH_F51	NE_F51	LONG RUN EMPHASIS(LRE)					
NH_F52	NE_F52	GRAY LEVEL NON-UNIFORMITY (GLN)					
NH_F53	NE_F53	RUN PERCENTAGE (RP)				van Griethuysen et. al., 2019	
NH_F54	NE_F54	RUN LENGTH NON-UNIFORMITY (RLN)					
NH_F55	NE_F55	LOW GRAY LEVEL RUN EMPHASIS (LGRE)					
NH_F56	NE_F56	HIGH GRAY LEVEL RUN EMPHASIS (HGRE)					
NH_F57	NE_F57	Heterogeneity	Features of chromatin conformation (by paper of Verbeek et al)	Matlab implementation	https://onlinelibrary.wiley.com/doi/pdf/10.1002/cyto.990070513		
NH_F58	NE_F58	Homogeneity					
NH_F59	NE_F59	Margination					
NH_F60	NE_F60	Clumping					
NH_F61	NE_F61	Condensation					
NH_F62	NE_F62	Closest Neighborhood Distance	Distance based features	Matlab implementation			
NH_F63	NE_F63	Average Distance to 5 Closest Neighbors		Matlab implementation			
NH_F64	NE_F64	Capacity Fractal Dimension (box counting)	Fractal dimension features	Matlab implementation	https://github.com/reuter-lab/fdim/tree/master/matlab		
NH_F65	NE_F65	Information Fractal Dimension					
NH_F66	NE_F66	Correlation Fractal Dimension					
NH_F67	NE_F67	Probability Fractal Dimension					
NH_F68	NE_F68	Mean Lacunarity over all box sizes					
NH_F69	NE_F69	Median Lacunarity over all box sizes					